

Testing, QA & AI Evaluation Strategy

ServiceGuard AI

1. Test pyramid

```
Unit tests
↓
Integration tests
↓
API contract tests
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End-to-end workflows
↓
Load / resilience tests
```

2. Java

- JUnit + Mockito.
- Spring Boot integration tests.
- Testcontainers PostgreSQL for realistic persistence tests.
- Security/RBAC tests.
- Idempotency and transaction tests.

3. Python

- pytest for tools/services.
- Feature-schema validation.
- Model inference regression tests.
- RAG retrieval tests.
- Agent graph/tool routing tests.

4. AI evaluation

- Groundedness — answer supported by retrieved/live evidence.
- Retrieval relevance — correct runbook chunks retrieved.
- Tool correctness — required tools selected.
- Refusal/uncertainty behavior when data is missing.
- Structured-output validity.
- Prompt-injection resistance tests.

5. Performance targets

- CRUD API latency measured under realistic concurrency.
- Telemetry ingestion throughput measured separately.
- AI requests bounded by timeout and concurrency limits.
- Database query plans checked for high-volume telemetry.

6. Failure scenarios

- Python unavailable.

- LLM timeout.
- Weather/dependency API unavailable.
- No runbook match.
- Duplicate telemetry.
- Duplicate incident.
- Database failover/restart.
- Expired/invalid JWT.
- Redis unavailable.